

Theorie
das sollte man können
Lukas Mengis

Contents

1	The rank theorem	3
2	Dual Maps	4
2.a	Lemma(18.a.7)	4
2.b	Lemma & def. (18.a.9)	4
2.c	Lemma (18.a.11)	5
3	The isomorphism theorem (18.c.5)	7
3.a	$\overline{T}$ is well defined.	7
3.b	$\overline{T}$ is linear	7
3.c	$\overline{T}$ is injective	7
3.d	$\overline{T}$ is surjective	8
3.e	The commutativity of the diagram	8
4	Determinant	9
4.a	Definition of sgn of a permutation	9
4.b	Lemma (20.b.1)	10
4.c	20.b.5	10
4.d	20.b.3	12
4.e	Determinants of endomorphisms	12
4.f	Recall	13
5	Eigenvalues and Eigenvectors	13
5.a	20.a.4	13
6	Trace	14
6.a	Lemma 21.a.10	14
6.b	Geometric and algebraic multiplicities of eigenvalues (21.b3/22.b4)	14
7	The orthogonal Complement	15
7.a	def 22.b.1/2	15
7.b	22.b.3	17
8	QR decomposition	18
8.a	Extension of QR decomposition	18
9	Dual spaces & inner products	20
9.a	23.a.1	20
10	The adjoint map	21
10.a	23.a.2	23

1 The rank theorem

Let

$$T : V \rightarrow W$$

be a linear map, where V is finite dim. Then

$$\dim(\ker(T)) + \dim(\operatorname{Im}(T)) = \dim(V)$$

Proof.

Write $n := \dim(V)$.

Let $u_1, \dots, u_k$ be a basis for $\ker(T)$

$$\operatorname{span}(u_1, \dots, u_k) = \ker(T)$$

Extend the basis of $\ker(T)$ to a basis for V

$$\operatorname{span}(u_1, \dots, u_k, v_1, \dots, v_{n-k})$$

We can write the image of T as followed

$$\operatorname{Im}(T) = \operatorname{span}(T(u_1), \dots, T(u_k), T(v_1), \dots, T(v_{n-k})) = \operatorname{span}(T(v_1), \dots, T(v_{n-k}))$$

We claim that $Tv_1, \dots, Tv_{n-k}$ form a basis for $\operatorname{Im}(T)$. This proves that $Tv_1, \dots, Tv_{n-k}$ spans $\operatorname{Im}(T)$, so it remains to show that those vectors are linear indep.

Let $a_1, \dots, a_{n-k} \in K$, and assume $\sum_{i=1}^{n-k} a_i Tv_i = 0$.

Since T is linear

$$\begin{aligned} T\left(\sum_{i=1}^{n-k} a_i v_i\right) &= \sum_{i=1}^{n-k} a_i Tv_i = 0 \\ \Rightarrow \sum_{i=1}^{n-k} a_i v_i &\in \ker(T) \end{aligned}$$

Note $u_1, \dots, u_k$ is a basis for $\ker(T)$

$$\begin{aligned} \Rightarrow \exists b_1, \dots, b_k \quad \text{such that } a_1 v_1 + \dots, a_{n-k} v_{n-k} &= b_1 u_1 + \dots + b_k u_k. \\ \Rightarrow a_1 v_1 + \dots, a_{n-k} v_{n-k} - b_1 u_1 - \dots - b_k u_k &= 0 \end{aligned}$$

But $u_1, \dots, u_k, v_1, \dots, v_{n-k}$ are linear independent (they form a basis)

$$\Rightarrow a_1 = \dots = a_{n-k} = b_1 = \dots = b_k = 0$$

This proves that

$$Tv_1, \dots, Tv_{n-k}$$

form a basis for $\operatorname{Im}(T)$

Conclusion

$$\begin{aligned} \dim(T) &= n - k = \dim(V) - \dim(\ker(T)) \\ \dim(\ker(T)) + \dim(\operatorname{Im}(T)) &= \dim(V) \end{aligned}$$

□

2 Dual Maps

2.a Lemma(18.a.7)

The elemnts

$$v_1^*, \dots, v_n^*$$

form a basis for V^*

We denote this basis by

$$B^* = (v_1^*, \dots, v_n^*)$$

Moreover

$$\forall l \in V^*, \quad l = \sum_{i=1}^n l(v_i) v_i^*.$$

Proof.

Let $l \in V^*$. We claim that

$$\begin{aligned} l &= \sum_{i=1}^n l(v_i) v_i^* \\ \Leftrightarrow l(v) &= \sum_{i=1}^n l(v_i) v_i^*(v) \end{aligned}$$

Indeed, l and its representative sum are linear maps $V \rightarrow K$, so it's enough to check that in the second equation holds for $v = v_1, v = v_2, \dots, v = v_n$, bc $v_1, \dots, v_n$ form a basis for V .

Set $v = v_k$

$$\sum_{i=1}^n l(v_i) v_i^*(v_k) = \sum_{i=1}^n l(v_i) \delta_{ik} = l(v_k).$$

The cleaim just proven shows that $v_1^*, \dots, v_n^*$ span V^* since

$$\dim(V^*) = \dim(V) = n,$$

it follows that $v_1^*, \dots, v_n^*$ form a basis for V^* .

□

2.b Lemma & def. (18.a.9)

Let V, W be vect. spaces oves K . Let

$$T : V \rightarrow W$$

be a linear map. Define a new map

$$\begin{aligned} T^* : W^* &\rightarrow V^*, \quad T^*(l) := l \circ T \quad \forall l \in W^* \\ (l : W &\rightarrow K) \end{aligned}$$

Then T^* is a linear map. It is called the dual map to T .

Proof.

Let $l \in W^*$, i.e. $l : W \rightarrow K$ linear.

Clearly $l \circ T : V \rightarrow K$ is a linear map (b.c. both T and l are), so $T(l) := l \circ T \in V^*$. This shows $T^* : W^* \rightarrow V^*$ has its target indeed V^* as claimed.

We'll show now that T^* is a linear map.

Let

$$l \in W^*, \quad \alpha \in K$$

We have that

$$\begin{aligned}
T^*(\alpha l) &\in V^* \\
\Rightarrow T^*(\alpha l)(v) &= ((\alpha l) \circ T)(v) \\
&= (\alpha l)(Tv) \\
&= \alpha l(T(v)) \\
&= \alpha(l \circ T)(v) \\
&= \alpha(T^*l)(v) \\
\Rightarrow T^*(\alpha l) &= \alpha T^*(l)
\end{aligned}$$

Similarly one shows

$$\begin{aligned}
T^*(l_1 + l_2)(v) &= ((l_1 + l_2) \circ T)(v) \\
&= (l_1 \circ T + l_2 \circ T)(v) \\
&= (l_1 \circ T)(v) + (l_2 \circ T)(v) \\
&= T^*(l_1)(v) + T^*(l_2)(v) \\
\Rightarrow T^*(l_1 + l_2) &= T^*(l_1) + T^*(l_2)
\end{aligned}$$

□

2.c Lemma (18.a.11)

Let

$$S : V \rightarrow W$$

be a linear map between two f.dim. Vect.spaces V & W over K . Let B be a basis for V and K a basis for W .

Consider

$$S^* : W^* \rightarrow V^*$$

and the bases K^*, B^* for W^*, V^* , respectively.

Then

$$[S^*]_{B^*}^{K^*} = [S_K^B]^T$$

Proof.

Write

$$\begin{aligned}
B &= (v_1, \dots, v_n), \quad K = (w_1, \dots, w_m) \\
A &:= [S]_K^B, \quad A = (a_{ij}).
\end{aligned}$$

A satisfies

$$Sv_j = \sum_{i=1}^n a_{ij} w_i$$

Now for $w^* \in K^*$ we have

$$\begin{aligned}
S^* w_j^* &= w_j^* \circ S \\
&= \sum_{i=1}^n (w_j^* \circ S)(v_i) v_i^* \\
&= \sum_{i=1}^n w_j^*(S(v_i)) v_i^* \\
&= \sum_{i=1}^n w_j^* \left(\sum_{k=1}^n a_{ki} w_k \right) v_i^* = \sum_{i=1}^n a_{ji} v_i^*
\end{aligned}$$

$\Rightarrow$ coordinates of $S^* w_j^*$ in the basis $B^* = (v_1^*, \dots, v_n^*)$ are

$$\begin{pmatrix} a_{j1} \\ \vdots \\ a_{jn} \end{pmatrix}$$

row j in A , represented as col in the matrix A^T

□

3 The isomorphism theorem (18.c.5)

Let V, W be vect.sp. over K and

$$T : v \rightarrow W$$

a linear map. Define a new map

$$\bar{T} : V/\ker(T) \rightarrow \text{Im}(T)$$

Let $x \in V/\ker(T)$. Choose a representative $v \in V$ for x .

$$x = [v]$$

Define

$$\bar{T}(x) := T(v)$$

Then

1. $\bar{T}$ well defined, and is linear map.
2. $\bar{T} : V/\ker(T) \rightarrow \text{Im}(T)$ is an isomorphism.
3. The following diagram commutes

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \pi \downarrow & & \uparrow i \\ V/\ker(T) & \xrightarrow{\bar{T}} & \text{Im}(T) \end{array}$$

Proof.

3.a $\bar{T}$ is well defined.

Suppose $x = [v] = [v']$, we need to show that $T(v) = T(v')$.

Indeed $[v] = [v']$ implies that

$$\begin{aligned} v - v' &\in \ker(T). \\ \Rightarrow T(v - v') &= 0 \\ \Rightarrow T(v) &= T(v') \end{aligned}$$

3.b $\bar{T}$ is linear

$$\begin{aligned} \bar{T}(\alpha[v]) &= \bar{T}([\alpha v]) \\ &= T(\alpha v) \\ &= \alpha T(v) = \alpha \bar{T}([v]) \end{aligned}$$

$$\begin{aligned} \bar{T}([v_1] + [v_2]) &= \bar{T}([v_1 + v_2]) \\ &= T(v_1 + v_2) \\ &= T(v_1) + T(v_2) = T([v_1]) + T([v_2]) \end{aligned}$$

3.c $\bar{T}$ is injective

Let $x \in \ker(\bar{T})$. Write x as $x = [v]$ for some $v \in V$.

Then

$$\bar{T}(x) = T(v) \Rightarrow v \in \ker(T) \Rightarrow x = [v] = 0_{v/\ker(T)}$$

3.d $\overline{T}$ is surjective

Let

$$\begin{aligned} w \in \text{Im}(T). \quad &\Rightarrow \exists v \in V \text{ such that } T(v) = w \\ &\Rightarrow \overline{[v]} = T(v) = w \end{aligned}$$

This shows that

$$\text{Im}(T) \subseteq \text{Im}(\overline{T}).$$

But by the def. of $\overline{T}$ we also have $\text{Im}(\overline{T}) \subseteq \text{Im}(T)$

$$\Rightarrow \text{Im}(\overline{T}) = \text{Im}(T).$$

3.e The commutativity of the diagram

This follows immediately from the definitions.

4 Determinant

4.a Definition of sgn of a permutation

Let

$$D : M_{n \times n}(K) \rightarrow K$$

be an n-linear alternating function

Denote

$$I = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} = \begin{pmatrix} -\varepsilon_1 - \\ \vdots \\ -\varepsilon_n - \end{pmatrix}$$

$B = (\varepsilon_1, \dots, \varepsilon_n)$ is the standard basis of K_{row}^n
clearly

$$\alpha_i = \sum_{j=1}^n A(i, j) \varepsilon_j \quad \forall 1 \leq i \leq n$$

$$\begin{aligned} \Rightarrow D(A) &= D(\alpha_1, \dots, \alpha_n) \\ &= D\left(\sum_{j=1}^n A(1, j) \varepsilon_j, \alpha_2, \dots, \alpha_n\right) \\ &= \sum_{j=1}^n A(1, j) \cdot D(\varepsilon_j, \alpha_2, \dots, \alpha_n) \\ &= \sum_{j,k} A(1, j) \cdot A(2, k) \cdot D(\varepsilon_j, \varepsilon_k, \dots, \alpha_n) \\ &= \sum_{k_1, \dots, k_n} \left(\prod_{j=1}^n A(j, k_j) \right) D(I) \end{aligned}$$

Using the assumption that D is alternating. $\Rightarrow D(\varepsilon_{k_1}, \dots, \varepsilon_{k_n}) = n$ whenever two indices $k_1, \dots, k_n$ are equal.

So we are interested only in ordered sequence in which:

$$\forall k_j \neq k_i \quad \forall i \neq j$$

This is a permutation of degree n.

We can think of it as a bijective function

$$\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$$

then the sequence

$$(k_1, \dots, k_n) = (\sigma(1), \dots, \sigma(n))$$

is a permutation as defined earlier.

We can write

$$D(A) = \sum_{\sigma \in S_n} A(1, \sigma(1)) \cdot A(2, \sigma(2)) \cdot \dots \cdot A(n, \sigma(n)) D(\varepsilon_{\sigma(1)}, \dots, \varepsilon_{\sigma(n)}) = \sum_{\sigma \in S_n} D(\varepsilon_{\sigma(1)}, \dots, \varepsilon_{\sigma(n)}) \prod_{i=1}^n A(i, \sigma(i))$$

Let σ be a permutation of degree n. Then one can pass from $(1, \dots, n)$ to $(\sigma(1), \dots, \sigma(n))$ by performing a finite sequence of interchanges of pairs (transposition).

More precisely, transposition is a permutation:

$\theta : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ for which $\exists i, j \in \{1, \dots, n\}, i \neq j$
such that $\theta(i) = \theta(j)$ and $\theta(k) = k \quad \forall k \neq i, j$

Now, every permutation can be written as a composition of transpositions

$$\sigma = \theta_1 \circ \dots \circ \theta_m$$

The transposition are not unique. There are many different way

$$\sigma = \theta_1 \circ \dots \circ \theta_m = \theta'_1 \circ \dots \circ \theta'_{m'}$$

To show there are many different ways we can look at examples

$$\sigma := (1, 2, 3) \rightarrow (3, 2, 1)$$

$$\sigma = t_{0,1} \circ t_{1,2} \circ t_{0,1} = t_{0,2}$$

4.b Lemma (20.b.1)

If a permutation is obtained from two different ways, one time by a sequence of m permutations, and another time by a sequence of m' transposition, the m and m' have the same parity

$$(-1)^m = (-1)^{m'}$$

The parity only depends on σ and not on the a particular way we performed a seq. of transpositions. Note

$$\text{sgn}(\sigma) := (-1)^m \quad \forall m$$

Proof

Fix a determinant funct.

$$E : M_{n \times n}(K) \rightarrow K$$

consider

$$E(e_{\sigma(1)}, \dots, e_{\sigma(n)})$$

If $(\sigma(1), \dots, \sigma(n))$ can be obtained from $(1, \dots, n)$ by a sequence of m transposition then the matrix

$$\begin{pmatrix} - & e_{\sigma(1)} & - \\ & \vdots & \\ - & e_{\sigma(n)} & - \end{pmatrix}$$

can be obtained from $I = \begin{pmatrix} - & e_1 & - \\ & \vdots & \\ - & e_n & - \end{pmatrix}$ by a sequence of interchanges of m pairs of rows.

$$\Rightarrow E(e_{\sigma(1)}, \dots, e_{\sigma(n)}) = (-1)^m E(e_1, \dots, e_n) = (-1)^m E(I) = (-1)^m \quad \forall m$$

$$\Rightarrow m = m'$$

$$\Rightarrow D(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n A(i, \sigma(i))$$

4.c 20.b.5

$$\forall A, B \in M_{n \times n}(K) \text{ we have } \det(A \cdot B) = \det(A) \cdot \det(B)$$

Proof

Fix a mat $B \in M_{n \times n}(K)$, and consider a funct.

$$D : M_{n \times n}(K) \rightarrow K$$

$$D(A) := \det(A \cdot B)$$

Claim D is n-linear and alternating.

Proof of claim

Write

$$A = \begin{pmatrix} - & \alpha_1 & - \\ & \vdots & \\ - & \alpha_n & - \end{pmatrix}$$

If we view α_i as an $1 \times n$ matrix then $\alpha_1 \dots B$ is also $1 \times n$, and in fact

$$A \cdot B = \begin{pmatrix} - & \alpha_1 \cdot B & - \\ & \vdots & \\ - & \alpha_n \cdot B & - \end{pmatrix}$$

So

$$D(A) = \det(\alpha_1, \dots, \alpha_n \cdot B)$$

Clearly $\forall$ two row vector

gefällt mir nicht mache einen anderen Beweis

Proof

Let $A, B, C \in M_{n \times n}(K)$, define

$$C = AB$$

using matrix multiplication

$$c_{i,j} = \sum_{k=1}^n a_{i,k} b_{k,j}$$

Therefore

$$\begin{aligned} \det(C) &= \det(AB) \\ &= \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n c_{i,\sigma(i)} \\ &= \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n \left(\sum_{k_i}^n a_{i,k_i} b_{k_i,\sigma(i)} \right) \\ &= \sum_{k_1, \dots, k_n}^n \prod_{i=1}^n a_{i,k_i} \underbrace{\sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n b_{k_i,\sigma(i)}}_{k_i=k_j \Rightarrow =0} \end{aligned}$$

If $k_i = k_j$ then the right side is $= 0$, therefore we can change to assuming permutation

$$\Rightarrow \dots = \left(\sum_{\pi \in S_n} \prod_{i=1}^n a_{i,\pi(i)} \right) \left(\sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n b_{\pi(i),\sigma(i)} \right)$$

Let $\sigma = \tau \circ \pi \in S_n$, with $\tau \in S_n$

$$\Rightarrow \dots = \left(\sum_{\pi \in S_n} \prod_{i=1}^n a_{i,\pi(i)} \right) \left(\sum_{\tau \circ \pi \in S_n} \text{sgn}(\tau \circ \pi) \prod_{i=1}^n b_{\pi(i),(\tau \circ \pi)(i)} \right)$$

Since π is a permutation, we may reindex the product, and since multiplication in K is commutative, the order of the factors does not matter. Note that $\text{sgn}(\pi \circ \tau) = \text{sgn}(\pi) \text{sgn}(\tau)$.

$$\Rightarrow \dots = \left(\sum_{\pi \in S_n} \text{sgn}(\pi) \prod_{i=1}^n a_{i,\pi(i)} \right) \left(\sum_{\tau \in S_n} \text{sgn}(\tau) \prod_{j=1}^n b_{j,\tau(j)} \right) = \det(A) \det(B)$$

(Since $\text{sgn}(\pi)$ is independent of τ , it can be factored out of the inner sum and absorbed into the outer sum over π .)

4.d 20.b.3

$\forall$ n -linear. alternating function

$$D : M_{n \times n}(K) \rightarrow K$$

Let $A \in M_{n \times n}(K)$

$$\Rightarrow \alpha_i = \sum_{j=1}^n A(i, j) e_j \quad \forall 1 \leq i \leq n$$

we have

$$\begin{aligned} D(A) &= D(\alpha_1, \dots, \alpha_n) \\ &= \sum_{i_1, \dots, i_n} \prod_{k=1}^n a(i_k, k) D(e_{i_1}, \dots, e_{i_n}) \end{aligned}$$

Since D is alternating every term with equal indices vanishes. We can switch to permutation.

$$\Rightarrow D(A) = \sum_{\sigma \in S_n} \prod_{k=1}^n a(\sigma(k), k) D(e_{\sigma(1)}, \dots, e_{\sigma(n)})$$

Since D is alternating

$$D(e_{\sigma(1)}, \dots, e_{\sigma(n)}) = \text{sgn}(\sigma) D(I)$$

therefore

$$\begin{aligned} D(A) &= \left(\sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a(\sigma(i), i) \right) D(I) \\ &= \det(A) D(I) \end{aligned}$$

4.e Determinants of endomorphisms

Let

$$A \in M_{n \times n}(K) \quad \text{and} \quad P \in GL(n, K)$$

Then

$$\begin{aligned} \det(P^{-1}AP) &= \det(P^{-1}) \det(A) \det(P) \\ &= \frac{1}{\det(P)} \det(A) \det(P) = \det(A) \end{aligned}$$

$\Rightarrow$ similar matrices have the same determinant.

Let V be a finite dim. vect space over K . Put $n := \dim(V)$. Let

$$T : V \rightarrow V \quad \text{a linear map}$$

Pick a basis B for V . This gets us a matrix $[T]_B^B \in M_{n \times n}(K)$.

Consider $\det([T]_B^B)$.

Does $\det([T]_B^B)$ depend on B ?

Let B' another basis for V . Denote by

$$P := [\text{id}_V]_B^{B'}$$

the transition mat. between the basis B' and B .

Recall that

$$[T]_{B'}^{B'} = [\text{id}_V]_{B'}^B \cdot [T]_B^B \cdot [\text{id}_V]_B^{B'}$$

and also we have

$$[\text{id}_V]_{B'}^B = ([\text{id}_V]_B^{B'})^{-1}$$

So

$$[T]_{B'}^{B'} = P^{-1}[T]_B^B P \Rightarrow \det([T]_{B'}^{B'}) = \det([T]_B^B)$$

4.f Recall

For linear maps $S, T : V \rightarrow V$ and a basis B :

$$[S \circ T]_B^B = [S]_B^B [T]_B^B$$

If T is an isomorphism,

$$[T^{-1}]_B^B = ([T]_B^B)^{-1}.$$

The identity map is represented by the identity matrix:

$$[\text{id}_V]_B^B = I.$$

5 Eigenvalues and Eigenvectors

5.a 20.a.4

Let $\lambda_1, \dots, \lambda_n$ be pairwise distinct eigenvalues of T with eigenvectors $v_1, \dots, v_n$.

We prove by induction on n .

Base case $n = 1$: since $v_1 \neq 0$, the set $\{v_1\}$ is linearly independent.

Induction step: Assume $v_1, \dots, v_n$ are linearly independent. Let

$$a_1 v_1 + \dots + a_n v_n + a_{n+1} v_{n+1} = 0. \quad (*)$$

Apply T :

$$a_1 \lambda_1 v_1 + \dots + a_n \lambda_n v_n + a_{n+1} \lambda_{n+1} v_{n+1} = 0. \quad (**)$$

Now compute $(**) - \lambda_{n+1} (*)$:

$$a_1 (\lambda_1 - \lambda_{n+1}) v_1 + \dots + a_n (\lambda_n - \lambda_{n+1}) v_n = 0.$$

By the induction hypothesis,

$$a_i (\lambda_i - \lambda_{n+1}) = 0 \quad \forall i = 1, \dots, n.$$

Since $\lambda_i \neq \lambda_{n+1}$,

$$a_1 = \dots = a_n = 0.$$

Substitute into $(*)$:

$$a_{n+1} v_{n+1} = 0.$$

Since $v_{n+1} \neq 0$,

$$a_{n+1} = 0.$$

Thus

$$a_1 = \dots = a_n = a_{n+1} = 0,$$

so $v_1, \dots, v_n, v_{n+1}$ are linearly independent.

6 Trace

6.a Lemma 21.a.10

$$\forall A, B \in M_{n \times n}(K)$$

$$\text{Tr}(A \cdot B) = \text{Tr}(B \cdot A)$$

If A and B are similar matrices then

$$\text{Tr}(A) = \text{Tr}(B)$$

In particular, if B and B' are two bases for V then

$$\text{Tr}([T]_B^B) = \text{Tr}([T]_{B'}^{B'})$$

Thus, $\text{Tr}(T)$ is well-defined.

Proof

1)

$$\begin{aligned} \text{Tr}(A \cdot B) &= \sum_{i=1}^n (A \cdot B)_{i,i} \\ &= \sum_{i=1}^n \sum_{j=1}^n A_{ij} B_{ji} \\ &= \sum_{i,j} A_{ij} B_{ji} \\ &= \sum_{i,j} A_{ji} B_{ij} \\ &= \sum_{i=1}^n \sum_{j=1}^n A_{ji} B_{ij} \\ &= \sum_{i=1}^n (B \cdot A)_{ii} = \text{Tr}(B \cdot A) \end{aligned}$$

2) If $B = P^{-1}AP$ then

$$\begin{aligned} \text{Tr}(B) &= \text{Tr}(P^{-1}AP) \\ &= \text{Tr}(P^{-1} \cdot (AP)) \\ &\stackrel{\text{by 1}}{=} \text{Tr}(APP^{-1}) = \text{Tr}(A) \end{aligned}$$

3)

$$\begin{aligned} \text{Tr}([T]_{B'}^{B'}) &= \text{Tr}([\text{id}_V]_{B'}^B \cdot [T]_B^B \cdot [\text{id}_V]_B^{B'}) \\ &\stackrel{\text{by 1}}{=} \text{Tr}([T]_B^B) \end{aligned}$$

6.b Geometric and algebraic multiplicities of eigenvalues (21.b3/22.b4)

Let λ be an eigenvalue of $T \in \text{End}(V)$.

$$m_g(T; \lambda) = \dim(E_T(\lambda)).$$

If V is finite-dimensional,

$$m_a(T; \lambda) = \text{multiplicity of } \lambda \text{ in } P_T(x).$$

Prob (21.b.4)

$$m_g(T; \lambda) \leq m_a(T; \lambda)$$

Proof

Let

$$k = m_{g(T; \lambda)} = \dim(E_{T(\lambda)}).$$

Choose a basis

$$(v_1, \dots, v_k)$$

of $E_{T(\lambda)}$ and extend it to a basis

$$B = (v_1, \dots, v_k, w_{k+1}, \dots, w_n)$$

of V .

Then

$$[T]_B^B = \begin{pmatrix} \begin{array}{c|c} & \\ \hline [Tv_1]_B & \dots & [Tv_k]_B \\ \hline & & \end{array} & \begin{array}{c} \\ \\ \\ \end{array} \end{pmatrix} = \begin{pmatrix} \lambda I_k & \star \\ 0 & C \end{pmatrix}$$

where $C \in M_{(n-k) \times (n-k)}(K)$

Hence

$$P_{T(x)} = \det([T]_B^B - xI_n) = (\lambda - x)^k \det(C - xI_{n-k}).$$

Therefore,

$$(\lambda - x)^k \mid P_T(x),$$

so λ is a root of $P_T(x)$ of multiplicity at least k :

$$m_a(T; \lambda) \geq k.$$

Since

$$k = m_g(T; \lambda),$$

we conclude

$$m_g(T; \lambda) = k \leq m_a(T; \lambda).$$

□

7 The orthogonal Complement

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space over $(\mathbb{R} \text{ or } \mathbb{C})$

7.a def 22.b.1/2

Let $\emptyset \neq S \subseteq V$ be a subset. The orthogonal complement on S is defined as followed.

$$S^\perp := \{v \in V \mid v \perp s \quad \forall s \in S\} = \{v \in V \mid \langle v, s \rangle = 0\}$$

When

$$S = \{v\} \quad \text{we write } v^\perp := \{v\}^\perp$$

proof

1) $S^\perp \subseteq V$ is a linear subspace

$$0 \in S^\perp \text{ b.c. } \langle 0, s \rangle = 0 \quad \forall s \in S$$

If $\alpha, \beta \in K, v, w \in S^\perp$ then $\forall s \in S$ we have

$$\langle \alpha v + \beta w, s \rangle = \alpha \langle v, s \rangle + \beta \langle w, s \rangle = 0 \Rightarrow \alpha v + \beta w \in S^\perp$$

$$2) 0^\perp = V, V^\perp = \{0\}$$

$$0^\perp = V$$

$$0^\perp = \{v \in V \mid \langle 0, v \rangle = 0\} = V$$

Let $v \in V^\perp$. $\Rightarrow \langle v, w \rangle = 0 \quad w \in V$. In particular

$$\langle v, v \rangle = 0 \Rightarrow v = 0$$

This shows $V^\perp \subseteq \{0\}$. But clearly $\{0\} \in V^\perp$. So $V^\perp = \{0\}$.

3) $S \cap S^\perp$ is either $\emptyset$ or equal to $\{0\}$

If $S \cap S^\perp = \emptyset$ we are done. otherwise, assume that $s \in S \cap S^\perp \Rightarrow \langle s, s \rangle = 0 \Rightarrow s = 0$

4) If $S \subseteq T$ then $S^\perp \supseteq T^\perp$

Let $s \in T^\perp$

$$\langle s, t \rangle = 0 \quad \forall t \in T$$

Since

$$S \subseteq T$$

this holds $\forall s \in S$.

so

$$T^\perp \subseteq S^\perp$$

5) $(\text{Sp}(S))^\perp = S^\perp$

We have $S \subseteq \text{span}(S)$.

By 4),

$$(\text{span}(S))^\perp \subseteq S^\perp.$$

We'll show that

$$S^\perp \subseteq (\text{span}(S))^\perp.$$

Indeed, let $v \in S^\perp$.

Let $u \in \text{span}(S)$. Write

$$u = \sum_{i=1}^k \alpha_i u_i$$

with $\alpha_i \in K$ and $u_i \in S$.

Then

$$\langle v, u \rangle = \sum_{i=1}^k \alpha_i \langle v, u_i \rangle = 0.$$

Hence

$$v \in (\text{span}(S))^\perp.$$

Therefore

$$(\text{span}(S))^\perp = S^\perp.$$

6) Let $s \in S$.

Then for all $v \in S^\perp$ we have

$$\langle v, s \rangle = 0.$$

Hence

$$\langle s, v \rangle = 0.$$

So

$$s \in (S^\perp)^\perp.$$

Thus

$$S \subseteq (S^\perp)^\perp.$$

7.b 22.b.3

Let V be an inner product space and $U \subseteq V$ a f.dim. subspace. Then $U^\perp$ is a complement of U in V ($U + U^\perp = V$ & $U \cap U^\perp = \{0\}$).

Proof

By the Lemma we have

$$U \cap U^\perp = \{0\}.$$

So it's enough to show that

$$U + U^\perp = V$$

Put

$$r := \dim(V)$$

and pick an orthonormal basis for U

$$B = (e_1, \dots, e_r)$$

Let $v \in V$.

Define

$$\tilde{P}_U(v) := \underbrace{\langle v, e_1 \rangle e_1 + \dots + \langle v, e_r \rangle e_r}_{\text{The orthogonal projection of } v \text{ to/on } U} \in U$$

We have

$$v \in \underbrace{\tilde{P}_U(v)}_{\in U} + (v - \tilde{P}_U(v)). \quad (\star)$$

We claim that

$$v - \tilde{P}_U(v) \in U^\perp$$

Indeed

$$\begin{aligned} \langle v - \tilde{P}_U(v), e_i \rangle &= \langle v, e_i \rangle - \sum_{k=1}^r \langle v, e_k \rangle \cdot \langle e_k, e_i \rangle \quad \text{recall } \tilde{P}_U(v) = \sum_{k=1}^r \langle v, e_k \rangle e_k \\ &= \langle v, e_i \rangle - \underbrace{\langle v, e_i \rangle}_{\text{due to the orthonormal basis}} = 0 \end{aligned}$$

This holds $\forall i$, hence

$$v - \tilde{P}_U(v) \perp \text{Sp}(e_1, \dots, e_r) = U \Rightarrow v - \tilde{P}_U(v) \in U^\perp$$

Going back to $(\star)$ we see that

$$v = \underbrace{\tilde{P}_U(v)}_{\in U} + \underbrace{(v - \tilde{P}_U(v))}_{\in U^\perp}.$$

This holds $\forall v \in V. \Rightarrow V = U + U^\perp$

□

Remark. If $U \subseteq V$ is a f.dim. subspace, then since $U^\perp$ is a complement of U in V , we have a ca.is.

$$V \cong U \oplus U^\perp.$$

8 QR decomposition

1) Let $A \in \text{GL}_n(\mathbb{R})$. Then $\exists Q \in O(n)$ and an upper triangular matrix $R \in M_{n \times n}(\mathbb{R})$ such that $A = Q \cdot R$.

2) Let $A \in \text{GL}_n(\mathbb{C})$. Then $\exists Q \in U(n)$ and an upper triangular matrix $R \in M_{n \times n}(\mathbb{C})$ such that $A = Q \cdot R$.

Proof (1/2 together).F Write

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix} \quad \text{with} \quad v_k \in K_{\text{col}}^n \quad (K = \mathbb{R} \text{ or } \mathbb{C})$$

Since $A \in \underbrace{\text{GL}_n(K)}_{\Rightarrow \text{full rank}}$, $v_1, \dots, v_n$ form a basis for K^n . Apply G-M orthogonalization to

$$v_1, \dots, v_n$$

and obtain an orthonormal basis

$$e_1, \dots, e_n \quad \text{for } K^n$$

Note that $\forall 1 \leq j \leq n$,

$$\text{Sp}(v_1, \dots, v_j) = \text{Sp}(e_1, \dots, e_j)$$

And we also have:

$$\begin{aligned} v_1 &= \langle v_1, e_1 \rangle e_1 \\ v_2 &= \langle v_2, e_1 \rangle e_1 + \langle v_2, e_2 \rangle e_2 \\ &\vdots \\ v_j &= \langle v_j, e_1 \rangle e_1 + \dots + \langle v_j, e_j \rangle e_j \\ &\vdots \\ v_n &= \langle v_n, e_1 \rangle e_1 + \dots + \dots + \langle v_n, e_n \rangle e_n. \end{aligned}$$

$$\Rightarrow \underbrace{\begin{pmatrix} | & & | & & | \\ v_1 & \dots & v_j & \dots & v_n \\ | & & | & & | \end{pmatrix}}_A = \underbrace{\begin{pmatrix} | & & | & & | \\ e_1 & \dots & e_i & \dots & e_n \\ | & & | & & | \end{pmatrix}}_Q \underbrace{\begin{pmatrix} \langle v_1, e_1 \rangle & \langle v_2, e_1 \rangle & \dots & \langle v_j, e_1 \rangle & \dots & \langle v_n, e_1 \rangle \\ 0 & \langle v_2, e_2 \rangle & \dots & \langle v_j, e_2 \rangle & \dots & \langle v_n, e_2 \rangle \\ \vdots & 0 & \ddots & \vdots & & \vdots \\ \vdots & \vdots & & \langle v_j, e_j \rangle & & \vdots \\ \vdots & \vdots & & 0 & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \dots & \langle v_n, e_n \rangle \end{pmatrix}}_R$$

□

8.a Extension of QR decomposition

Let $A \in M_{n \times n}(K)$ and let $r := \text{rank}(A)$. Write

$$A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_n \\ | & | & & | \end{pmatrix}, \quad v_j \in K^n.$$

If $r = 0$, then $A = 0$. Hence we may take $R = 0$ and Q arbitrary orthogonal/unitary.

Assume now that $r \geq 1$. Choose indices

$$1 \leq i_1 < \dots < i_r \leq n$$

inductively as follows: i_1 is the minimal index such that $v_{i_1} \neq 0$. If $i_1, \dots, i_k$ are already chosen, then i_{k+1} is the minimal index $l > i_k$ such that

$$v_l \notin \text{span}(v_{i_1}, \dots, v_{i_k}).$$

The construction stops after r steps. Thus, for every $k = 1, \dots, r$,

$$v_{i_1}, \dots, v_{i_k}$$

is a basis of

$$\text{span}(v_1, \dots, v_{i_k}).$$

Apply Gram–Schmidt to

$$v_{i_1}, \dots, v_{i_r}$$

and obtain an orthonormal system

$$e_1, \dots, e_r$$

such that for every $k = 1, \dots, r$,

$$\text{span}(e_1, \dots, e_k) = \text{span}(v_{i_1}, \dots, v_{i_k}) = \text{span}(v_1, \dots, v_{i_k}).$$

Extend $e_1, \dots, e_r$ to an orthonormal basis

$$e_1, \dots, e_r, e_{r+1}, \dots, e_n$$

of K^n .

Define

$$Q = \begin{pmatrix} | & | & & | \\ e_1 & e_2 & \dots & e_n \\ | & | & & | \end{pmatrix}.$$

Then

$$Q \in O(n) \text{ if } K = \mathbb{R}, \quad Q \in U(n) \text{ if } K = \mathbb{C}.$$

Since every column v_j belongs to $\text{Im}(A)$, we have

$$v_j = \sum_{k=1}^r \langle v_j, e_k \rangle e_k.$$

Also,

$$\langle v_j, e_k \rangle = 0 \quad \text{for all } k > r.$$

Moreover, by the minimal choice of the indices, if

$$i_k \leq j < i_{k+1},$$

then

$$v_j \in \text{span}(v_1, \dots, v_{i_k}) = \text{span}(e_1, \dots, e_k).$$

Hence

$$\langle v_j, e_l \rangle = 0 \quad \text{for all } l > k.$$

Thus

$$A = QR,$$

where

$$R = \begin{pmatrix} \langle v_1, e_1 \rangle & \langle v_2, e_1 \rangle & \cdots & \langle v_n, e_1 \rangle \\ \langle v_1, e_2 \rangle & \langle v_2, e_2 \rangle & \cdots & \langle v_n, e_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle v_1, e_r \rangle & \langle v_2, e_r \rangle & \cdots & \langle v_n, e_r \rangle \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix} \in M_{n \times n}(K).$$

The last $n - r$ rows of R are zero. Moreover, the nonzero rows have the same triangular structure as in the full-rank case, with pivots in the columns

$$i_1, \dots, i_r.$$

Therefore R has the block form

$$R = \begin{pmatrix} C & * \\ 0 & 0 \end{pmatrix},$$

where $C \in M_{r \times r}(K)$ is upper triangular after restricting to the pivot columns $i_1, \dots, i_r$.

Hence

$$A = QR$$

Zum Verständnis. Die Diagonale der C Matrix muss nicht immer ungleich null sein nur solange die Spaltenvektoren von in Serie pairwise distinct sind, wenn auf einmal ein Spaltenvektor kommt der abhängig von den vorderen Spalten ist dann, macht es wie einen Horizontalen Stritt nach rechts (pivot columns $i_1, \dots, i_r$).

9 Dual spaces & inner products

Assume $v \in K^n$ and let

$$\varphi : K^n \rightarrow K$$

be a functional. Then $\exists!$

$$A = (a_1, \dots, a_n) \in M_{1 \times n}(K) \text{ such that } \varphi(v) = A \cdot v \quad \forall v \in K^n.$$

Indeed, let $e_1, \dots, e_n$ be the std. basis for K^n and define

$$a_1 = \varphi(e_1), \dots, a_n = \varphi(e_n)$$

Let

$$v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in K^n$$

Then

$$\varphi(v) = \varphi \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \varphi(v_1 e_1 + \dots + v_n e_n) = v_1 \varphi(e_1) + \dots + v_n \varphi(e_n)$$

$$v_1 a_1 + \dots + v_n a_n = A \cdot v.$$

9.a 23.a.1

Let $(V, \langle \cdot, \cdot \rangle)$ be f.dim. inner prod.sp. over K ($\mathbb{R}$ or $\mathbb{C}$). Let $\varphi \in V^*$.

Then

$$\exists! u \in V \text{ such that } \varphi = \varphi_u \text{ id est } \varphi(v) = \langle v, u \rangle \quad \forall v \in V$$

In fact for $K = \mathbb{R}$ the map

$$\begin{aligned}\Phi : V &\rightarrow V^* \\ v &\mapsto \varphi_v\end{aligned}$$

is an isomorphism. For $K = \mathbb{C}$, this map is injective & surjective but only $\mathbb{C}$ -antilinear

$$\begin{aligned}\text{id est } \Phi(u_1 + u_2) &= \Phi(u_1) + \Phi(u_2), \quad \forall u_1, u_2 \in V \\ \Phi(c \cdot u) &= \bar{c} \cdot \Phi(u), \quad \forall c \in \mathbb{C}, \quad u \in V\end{aligned}$$

Proof (that $\forall \varphi \in V^* \quad \exists! u$ such that $\varphi = \varphi_u$) Fix an orthonormal basis

$$e_1, \dots, e_n$$

for V ($n := \dim(V)$)

Let $\varphi \in V^*$. Let $v \in V$. We have

$$\begin{aligned}v &= \sum_{i=1}^n \langle v, e_i \rangle e_i \\ \Rightarrow \varphi(v) &= \varphi\left(\sum_{i=1}^n \langle v, e_i \rangle e_i\right) \\ &= \sum_{i=1}^n \langle v, e_i \rangle \varphi(e_i) \\ &= \langle v, \sum_{i=1}^n \overline{\varphi(e_i)} \cdot e_i \rangle\end{aligned}$$

Define

$$u = \sum_{i=1}^n \overline{\varphi(e_i)} \cdot e_i \Rightarrow \varphi = \varphi_u$$

Uniqueness of u. Suppose that $\varphi_{u_1} = \varphi_{u_2}$ for $u_1, u_2 \in V$

$$\Rightarrow \langle v, u_1 \rangle = \langle v, u_2 \rangle \quad \forall v \in V.$$

(go through the standard basis to check)

linearity obvious

10 The adjoint map

Let V, W be vect. spaces over K .

Let

$$T : V \rightarrow W$$

be a linear map. This induces a linear map

$$T^* : W^* \rightarrow V^*$$

by

$$T^*(\varphi) := \varphi \circ T$$

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ & \searrow \varphi \circ T & \downarrow \varphi \\ & & K \end{array}$$

Assume that

$$(V, \langle \cdot, \cdot \rangle_V) \quad (W, \langle \cdot, \cdot \rangle_W)$$

are inner product spaces over K .

Assume $\dim(V) < \infty$.

Given T , we can define

$$T' : W \rightarrow V$$

using “canonical” identifications

$$W \simeq W^* \quad V \simeq V^*.$$

$$\begin{array}{ccc}
 W^* & \xrightarrow{T^*} & V^* \\
 \uparrow \Phi_W & & \downarrow \Phi_V^{-1} \\
 W & \xrightarrow{\quad T' \quad} & V
 \end{array}$$

id est $T' := \Phi_V^{-1} \circ T^* \circ \Phi_W$
here we're using that $\dim V < \infty$)

Note that T' is uniquely defined by

$$\begin{aligned}
 \Phi_V \circ T' &= T^* \circ \Phi_W \\
 \Leftrightarrow \forall w \in W, \quad \Phi_V \circ T'(w) &= T^* \Phi_W(w) \\
 \Leftrightarrow \varphi_{T'(w)} &= T^* \varphi_w (\in V^*) \\
 \Leftrightarrow \forall v \in V, \quad \varphi_{T'(w)}(v) &= (T^* \varphi_w)(v) \\
 \Leftrightarrow \langle v, T'(w) \rangle_V &= \varphi_w(T(v)) = \langle T(v), w \rangle_W
 \end{aligned}$$

In other words,

$$\langle Tv, w \rangle_W = \langle v, T'w \rangle_V \quad \forall v \in V, w \in W$$

Claim T' is a linear map.

Proof. For $K = \mathbb{R}$, Φ_W, Φ_V and T^* are linear $\Rightarrow T'$ too.

Define

$$T' := \Phi_V^{-1} \circ T^* \circ \Phi_W : W \rightarrow V.$$

For $K = \mathbb{R}$ all maps are linear, hence T' is linear.

Now let $K = \mathbb{C}$. The maps Φ_W, Φ_V^{-1} are additive and conjugate-linear, while T^* is linear.

Additivity:

$$\begin{aligned}
 T'(w_1 + w_2) &= \Phi_V^{-1}(T^*(\Phi_W(w_1 + w_2))) \\
 &= \Phi_V^{-1}(T^*(\Phi_W(w_1) + \Phi_W(w_2))) \\
 &= \Phi_V^{-1}(T^*\Phi_W(w_1) + T^*\Phi_W(w_2)) \\
 &= T'(w_1) + T'(w_2).
 \end{aligned}$$

Homogeneity: for $c \in \mathbb{C}$,

$$\begin{aligned}
T'(cw) &= \Phi_V^{-1}(T^*(\Phi_W(cw))) \\
&= \Phi_V^{-1}(T^*(\bar{c}\Phi_W(w))) \\
&= \Phi_V^{-1}(\bar{c}T^*(\Phi_W(w))) \\
&= c\Phi_V^{-1}(T^*(\Phi_W(w))) \\
&= cT'(w).
\end{aligned}$$

Thus T' is K -linear.

10.a 23.a.2

Let V, W, U be finite-dimensional inner product spaces over $K \in \{\mathbb{R}, \mathbb{C}\}$.

Let $S, T : V \rightarrow W$ and $R : W \rightarrow U$ be linear maps. Then:

- 1) T^* is linear,
- 2) $(S + T)^* = S^* + T^*$,
- 3) $(\lambda T)^* = \bar{\lambda}T^*$, $\forall \lambda \in K$,
- 4) $(T^*)^* = T$,
- 5) $(\text{id}_V)^* = \text{id}_V$,
- 6) $(R \circ T)^* = T^* \circ R^*$.

1) Was proved already.

2) Let $w \in W, v \in V$.

$$\begin{aligned}
\langle v, (S + T)^*w \rangle &= \langle (S + T)v, w \rangle \\
&= \langle Sv, w \rangle + \langle Tv, w \rangle \\
&= \langle v, S^*w \rangle + \langle v, T^*w \rangle \\
&= \langle v, (S^* + T^*)w \rangle.
\end{aligned}$$

Thus $(S + T)^*w = (S^* + T^*)w$, hence $(S + T)^* = S^* + T^*$.

3) Let $\lambda \in K, v \in V, w \in W$.

$$\begin{aligned}
\langle v, (\lambda T)^*w \rangle &= \langle (\lambda T)v, w \rangle \\
&= \langle \lambda Tv, w \rangle \\
&= \lambda \langle Tv, w \rangle \\
&= \lambda \langle v, T^*w \rangle \\
&= \langle v, \bar{\lambda}T^*w \rangle.
\end{aligned}$$

Thus $(\lambda T)^*w = \bar{\lambda}T^*w$, hence $(\lambda T)^* = \bar{\lambda}T^*$.

4) Let $v \in V, w \in W$.

$$\begin{aligned}
\langle w, (T^*)^*v \rangle &= \langle T^*w, v \rangle \\
&= \overline{\langle v, T^*w \rangle} \\
&= \overline{\langle Tv, w \rangle} \\
&= \langle w, Tv \rangle.
\end{aligned}$$

Thus $(T^*)^*v = Tv$, hence $(T^*)^* = T$.

5) Let $v, w \in V$.

$$\begin{aligned}
\langle v, (\text{id}_V)^* w \rangle &= \langle \text{id}_{V(v)}, w \rangle \\
&= \langle v, w \rangle \\
&= \langle v, \text{id}_{V(w)} \rangle.
\end{aligned}$$

Thus $(\text{id}_V)^* w = \text{id}_{V(w)}$, hence $(\text{id}_V)^* = \text{id}_V$.

6) Let $v \in V, u \in U$.

$$\begin{aligned}
\langle v, (R \circ T)^* u \rangle &= \langle (R \circ T)v, u \rangle \\
&= \langle R(Tv), u \rangle \\
&= \langle Tv, R^* u \rangle \\
&= \langle v, T^*(R^* u) \rangle \\
&= \langle v, (T^* \circ R^*)u \rangle.
\end{aligned}$$

Thus $(R \circ T)^* u = (T^* \circ R^*)u$, hence $(R \circ T)^* = T^* \circ R^*$.